Definite integrals

1 Calculate the following definite integrals:

a $\int_{-4}^{2} e^{x} dx$

b $\int_{0}^{3} e^{4x} dx$

c $\int_{0}^{2} e^{\frac{3}{2}x} dx$

d $\int_{-2}^{4} e^{3x} dx$

e $\int_{-3}^{5} e^{-2x} dx$

f $\int_{0}^{2} (7+e^{x})\, dx$

g $\int_{0}^{3} \left(2e^{5x}+e^{-7x}\right) dx$

h $\int_{-3}^{3} \left(e^{\frac{1}{4}\theta}-3e^{-0.2\theta}\right) d\theta$

i $\int_{3}^{4} \left(e^{2x}+4\right)^{2} dx$

j $\int_{-4}^{4} \left(e^{x}+e^{2x}\right)^{2} dx$

k $\int_{-4}^{3} \dfrac{e^{-2x}+e^{3x}}{e^{x}} dx$

l $\int_{1}^{16} \left(e^{4x}+x^{2}-\sqrt{x}\right) dx$

2 Calculate the following definite integrals:

a $\int_{\frac{3\pi}{2}}^{2\pi} \cos x \; dx$

b $\int_{\frac{\pi}{2}}^{\pi} (-\sin x)\, dx$

c $\int_{\frac{\pi}{6}}^{\frac{7\pi}{6}} 2\sin x \; dx$

d $\int_{\frac{\pi}{6}}^{\frac{7\pi}{6}} 2\cos x \; dx$

e $\int_{-\frac{\pi}{6}}^{\frac{7\pi}{6}} \sin 3x \; dx$

f $\int_{-\frac{\pi}{6}}^{\frac{7\pi}{6}} \cos 3x \; dx$

g $\int_{-\frac{2\pi}{3}}^{2\pi} \sin\left(\frac{x}{4}\right) dx$

h $\int_{-\frac{4\pi}{3}}^{\frac{2\pi}{3}} \cos\left(\frac{x}{4}\right) dx$

i $\int_{-6}^{9} \sin\left(\frac{\pi x}{3}\right) dx$

j $\int_{-8}^{4} \cos\left(\frac{\pi x}{2}\right) dx$

k $\int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \sin(x+\pi)\, dx$

l $\int_{-\frac{\pi}{2}}^{\pi} \cos\left(4x-\frac{\pi}{2}\right) dx$

m $\int_{0}^{\frac{\pi}{6}} (\sin x+\cos x)\, dx$

n $\int_{-\frac{\pi}{6}}^{\frac{\pi}{6}} (4\cos x+\cos 4x)\, dx$

o $\int_{-\pi}^{\pi} \left(\sin\left(\frac{\theta}{6}\right)-\cos\left(\frac{\theta}{6}\right)\right) d\theta$

p $\int_{0}^{\frac{\pi}{3}} \left(2\cos 3x+\frac{1}{4}\sin 2x\right) dx$

q $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (8\cos(2x)-9\sin(3x))\, dx$

3 Consider the function $y=e^{x}+e^{-x}$.

a Find y'.

b Find the exact value of $\int_{0}^{3} \dfrac{e^{x}-e^{-x}}{e^{x}+e^{-x}} dx$.

4 Consider the function $y=e^{3x}\left(x-\dfrac{1}{3}\right)$.

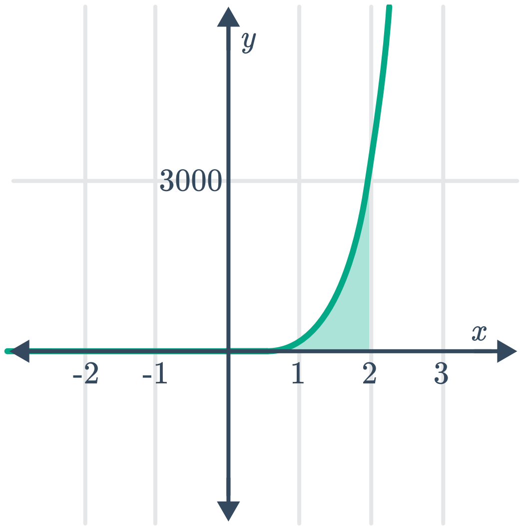
a Find y'.

b Hence find the exact value of $\int_{6}^{9} xe^{3x} dx$.

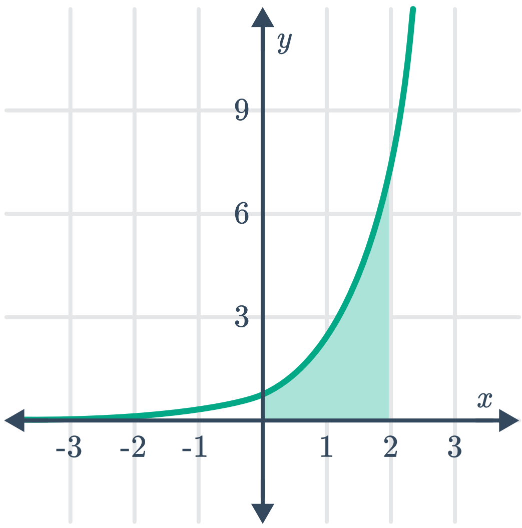
Graph symmetry

5 Calculate the exact area of the region enclosed between the following:

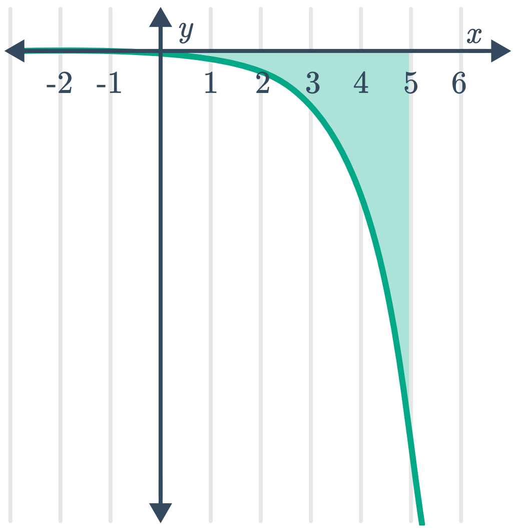
a $y = e^{4x}$, the $x-$axis, and the lines $x = -1$ and $x = 2$.



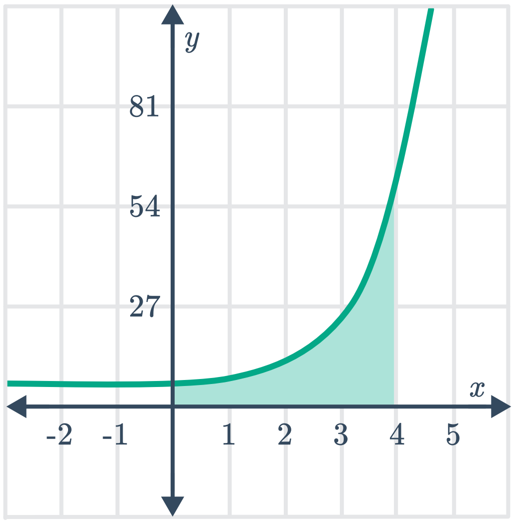
b $y = e^{x}$, the coordinate axes and the line $x = 2$.



c $y = -e^{x}$, the coordinate axes and the line $x = 5$.



d $y = e^x + 4$, the coordinate axes and the line $x = 4$.



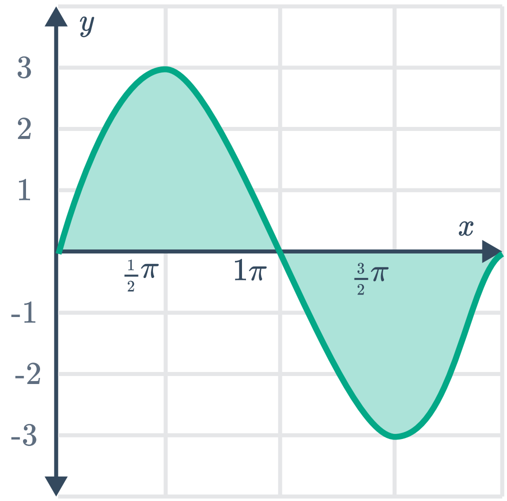
e $y = e^{-6x}$, the x-axis, the lines $x = -2$ and $x = 1$.

f $y = 6e^{3x}$, the x-axis, the lines $x = -3$ and $x = 3$.

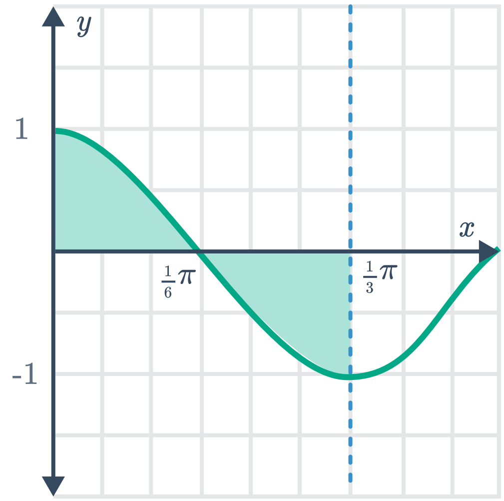
g $y = e^{\frac{x}{4}} + e^{-\frac{x}{4}}$, the coordinate axes and the line $x = \dfrac{1}{2}$.

6 Calculate the exact area of the regions bo[redacted]d by the x–axis and the curve:

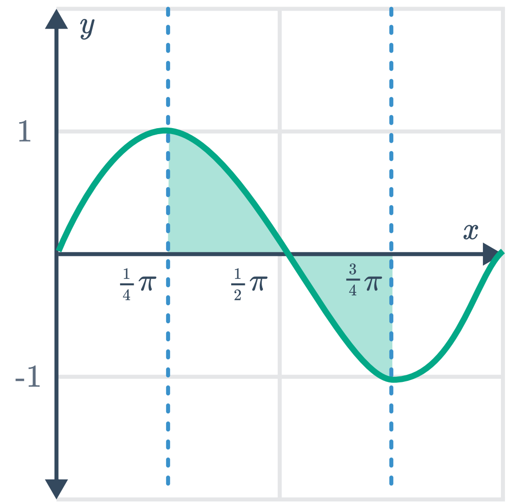
a $y = 3\sin x$, from $x = 0$ and $x = \pi$.



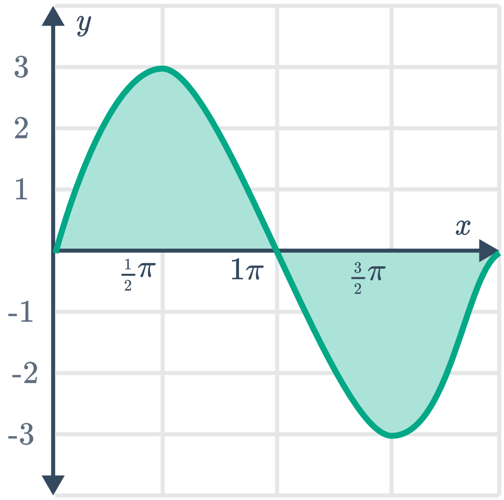
b $y = \cos 3x$, from $x = 0$ to $x = \dfrac{\pi}{3}$.



c $y = \sin 2x$, from $x = \dfrac{\pi}{4}$ to $x = \dfrac{3\pi}{4}$.



d $y = 3\sin x$, from $x = 0$ to $x = 2\pi$.



e $y = \sin x$, from $x = 0$ to $x = \dfrac{\pi}{2}$.

f $y = 3\cos x$, from $x = 0$ to $x = \dfrac{\pi}{2}$.

g $y = \cos x$, from $x = 0$ to $x = 2\pi$.

h $y = 3(\sin x + 1)$, from $x = 0$ to $x = 2\pi$.

i $y = \dfrac{1}{3} + 3\cos x$, from $x = -\dfrac{\pi}{6}$ to $x = \dfrac{\pi}{6}$.

j $y = 2 + \dfrac{1}{4}\sin 2x$, from $x = 0$ to $x = \dfrac{\pi}{2}$.

k $y = 8\cos 2x + 4\sin 2x$, from $x = 0$ to $x = \dfrac{\pi}{8}$.

7 Consider the function $f(x) = e^{3x} - e^{-3x}$.

a Find the x-intercept(s) of the function.

b State the limiting function as x approaches $+\infty$.

c Determine whether the function is odd or even.

d Find the exact area enclosed between the curve, the coordinate axes, and the line $x = 3$.

e Find the exact area bound by the curve, the x-axis, and the lines $x = 1$ and $x = -1$.

8 Consider the integral $\int_{-\frac{\pi}{12}}^{\frac{\pi}{12}} \sin 4x \, dx$.

a Evaluate the definite integral.

b State the values of x on the domain $-\frac{\pi}{12} \leq x \leq \frac{\pi}{12}$, for where the curve $y = \sin 4x$ is:

i Above the x-axis

ii Below the x-axis

c Hence calculate the exact area bound by the curve $y = \sin 4x$, the x-axis, and the lines $x = -\frac{\pi}{12}$ and $x = \frac{\pi}{12}$.

9 Consider the curve $y = \cos\left(\frac{x}{4}\right)$.

a Find the area bound by the curve and the axes, between $x = 0$ and $x = \frac{\pi}{2}$.

b Hence find the area bound by the curve and the x-axis, between $x = -4\pi$ and $x = 4\pi$.

10 Consider the curve $y = 4\sin\left(\frac{\pi x}{5}\right)$.

a State the period of the curve.

b Find the exact area bounded by the curve and the x-axis, between $x = \frac{5}{4}$ and $x = \frac{5}{2}$.

11 Consider the curve $y = \sin 3x$.

a Graph the function on the domain $0 \leq x \leq 2\pi$.

b Find the exact area of the region bounded by the curve, the x-axis and the lines $x = 0$ and $x = \frac{\pi}{4}$.

12 Consider the curve $y = 3\sin 2x$.

a Graph the function on the domain $0 \leq x \leq 2\pi$.

b Find the exact area bound by the curve and the x-axis between:

i $x = 0$ and $x = \dfrac{\pi}{2}$

ii $x = \dfrac{\pi}{4}$ and $x = \dfrac{2\pi}{3}$

13 Consider the function $y = 2 - \cos x$.

a Graph the function on the domain $0 \leq x \leq 2\pi$.

b Hence find the area represented by the integral $\int_0^{\frac{\pi}{2}} (2 - \cos x)\, dx$.

Use technology

14 Use technology to evaluate $\int_{-1}^{4} \dfrac{e^x}{\sqrt{x+4}}\, dx$. Round your answer to two decimal places.

15 Use technology to calculate the area bounded by the curve $y = e^{2x}\cos 3x$, the x-axis, and the lines $x = -\dfrac{\pi}{2}$ and $x = \dfrac{\pi}{6}$. Round your answer to two decimal places.